



RAJALAKSHMI INSTITUTE OF TECHNOLOGY
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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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SEMESTER VI

CB23531 – BUSINESS ANALYTICS LABORATORY

MINI PROJECT REPORT

REGISTER NUMBER	2117240030143
NAME	SIRI CHAND KANDHERI CHANDRABABU
PROJECT TITLE	Product return and reverse logistics analysis using return data
DATE OF SUBMISSION	
FACULTY IN-CHARGE	Ms. PANDIMA DEVI

Signature of Faculty In-charge

Signature of HoD

1. Project Title

Product return and reverse logistics analysis using return data (reason, time since purchase, channel) to model likelihood of return and suggest design or policy changes.

2. Introduction

Product returns are an important challenge for modern businesses, particularly in e-commerce and omnichannel retail. Although returns improve customer satisfaction and reduce purchase risk, they also create significant operational costs related to transportation, inspection, repackaging, restocking, refurbishment, replacement, and disposal.

This project analyzes product return data to understand the major factors associated with product returns. Key variables such as **return reason, time since purchase, sales channel, product characteristics, and customer/order information** can be analyzed to identify patterns in return behavior.

The project also develops a predictive model to estimate the likelihood of a product being returned. The results can help businesses reduce avoidable returns, improve product design and descriptions, optimize reverse-logistics processes, and develop more effective return policies.

3. Problem Statement

Businesses face increasing costs due to product returns and reverse logistics. However, without systematic analysis, it can be difficult to determine which products, channels, return reasons, or purchase patterns contribute most strongly to returns.

The problem addressed in this project is:

How can return data be analyzed and modeled to predict the likelihood of product returns and identify product, channel, and policy changes that can reduce return rates and reverse-logistics costs?

4. Aim / Goal

The main aim of the project is to analyze product return behavior and develop a predictive analytics framework that estimates return likelihood while generating actionable recommendations for product design, sales channels, and return policies.

5. Objectives

- Analyze historical product return data.
- Identify the most common reasons for product returns.
- Examine the relationship between time since purchase and return behavior.
- Compare return patterns across different sales channels.
- Identify product or transaction characteristics associated with high return rates.
- Develop a predictive model for return likelihood.
- Evaluate the performance of the predictive model.
- Identify business factors contributing to reverse-logistics activity.
- Recommend product-design improvements to reduce avoidable returns.
- Recommend return-policy and operational changes.
- Develop a dashboard or analytical reporting framework for monitoring return performance.

6. Business Context / Background

Product returns affect several areas of a business simultaneously. A returned product may require reverse transportation, quality inspection, repackaging, inventory updates, refurbishment, resale, replacement, or disposal.

High return rates can therefore reduce profit margins and create additional environmental and operational costs.

Common return reasons may include:

- Product damaged or defective
- Wrong size or fit
- Product not as expected
- Incorrect product shipped
- Customer changed their mind
- Late delivery
- Product quality concerns
- Duplicate purchase
- Other customer-specific reasons

Different sales channels may also exhibit different return patterns. For example, online purchases may experience higher returns because customers cannot physically inspect products before purchasing.

Analyzing these patterns allows businesses to move from simply processing returns toward **proactively preventing avoidable returns**.

7. Dataset Description

Data Source

Dataset: Synthetic E-Commerce Returns Management Dataset

Source: Kaggle

Link: <https://www.kaggle.com/datasets/sowmihari/returns-management>

The dataset is synthetic and designed to represent realistic e-commerce return-management scenarios.

Data Size

- **Records:** 5,000
- **Variables:** 23

Dataset Features (Columns)

Order_ID → Unique order identifier.

Product_ID → Unique product identifier.

User_ID → Unique customer identifier.

Order_Date → Date when the order was placed.

Return_Date → Date when the product was returned (if returned).

Product_Category → Category of the product (e.g., Clothing, Electronics, Books, Toys, etc.).

Product_Price → Price of the product per unit.

Order_Quantity → Number of units purchased in the order.

Discount_Applied → Discount percentage applied on the product.

Return_Status → Whether the order was Returned or Not Returned.

Return_Reason → Reason for return (e.g., Damaged, Wrong Item, Changed Mind).

Days_to_Return → Number of days taken by customer to return (0 if not returned).

User_Age → Age of the customer.

User_Gender → Gender of the customer (Male/Female).

User_Location → City/region of the customer.

Payment_Method → Mode of payment (Credit Card, Debit Card, PayPal, Gift Card, etc.).

Shipping_Method → Chosen shipping type (Standard, Express, Next-Day).

Return_Cost → Estimated logistics cost incurred when a return happens.

Profit_Loss → Net profit or loss for the order, considering product price, discount, and return cost.

CO2_Saved → Estimated CO₂ emissions saved (if return avoided).

Waste_Avoided → Estimated physical waste avoided (in units/items).

Target Variable

Return_Status is the target variable.

- 1 = Returned
- 0 = Not Returned

It is used to predict the likelihood of a product being returned.

8. Data Collection

The dataset will be downloaded from Kaggle and imported into Python.

The data-collection process consists of:

1. Downloading the CSV dataset from Kaggle.
2. Loading the dataset using Pandas.
3. Inspecting the structure and data types.
4. Checking for duplicate records.
5. Checking missing values.
6. Validating date fields.
7. Creating additional analytical variables where necessary.

9. Data Preprocessing

Data Cleaning

The dataset will be examined for:

- Duplicate orders.
- Duplicate product records.
- Invalid dates.
- Incorrect numerical values.
- Inconsistent categorical labels.

- Impossible return durations.
- Missing values.

Categorical values will be standardized where necessary.

Missing Value Handling

Missing values will be examined using:

- Missing-value counts.
- Missing-value percentages.
- Variable-level analysis.

For example, **Return_Date**, **Return_Reason**, and **Days_to_Return** may naturally be unavailable for products that were not returned.

Such missing values should not automatically be treated as errors because they may represent meaningful business information.

Outlier Detection

Numerical variables such as:

- Product Price
- Order Quantity
- Discount
- Days to Return
- Return Cost
- Profit/Loss

will be examined for outliers using box plots and the Interquartile Range method.

Extreme values will be investigated before deciding whether they should be retained, transformed, or removed.

Data Transformation

A useful derived variable will be:

Return Duration = Return Date – Order Date

If **Days_to_Return** is already available, it can be validated against the date fields.

Additional variables may include:

- Order Month

- Order Year
- Return Month
- Product Price Category
- Discount Category
- Return Cost Category

Data Encoding / Normalization

Categorical variables such as:

- Product Category
- Return Reason
- Payment Method
- Shipping Method
- User Gender
- User Location

will be encoded using techniques such as one-hot encoding.

Numerical features can be standardized when required by the selected machine-learning algorithm.

10. Exploratory Data Analysis (EDA)

Exploratory Data Analysis is performed to understand patterns and relationships within the dataset.

10.1 Overall Return Rate

The overall return rate is calculated as:

$$\text{Return Rate} = \text{Returned Orders} / \text{Total Orders} \times 100$$

This provides a basic measure of the return problem.

10.2 Return Reasons

The frequency of each return reason is analyzed.

A bar chart is used to identify the most common reasons.

This helps the business determine which causes should be prioritized for corrective action.

10.3 Return Rate by Product Category

Return rates are compared across product categories.

Categories with unusually high return rates may require:

- Product redesign.
- Better quality control.
- Improved product descriptions.
- Better sizing information.
- Supplier investigation.

10.4 Days to Return

The distribution of **Days_to_Return** is analyzed.

This identifies whether customers tend to return products shortly after purchase or after longer periods.

10.5 Discount Analysis

The relationship between discounts and return behavior is analyzed.

This helps determine whether highly discounted products have significantly different return patterns.

10.6 Shipping Method Analysis

Return rates are compared across shipping methods.

This can help identify potential relationships between logistics processes and returns.

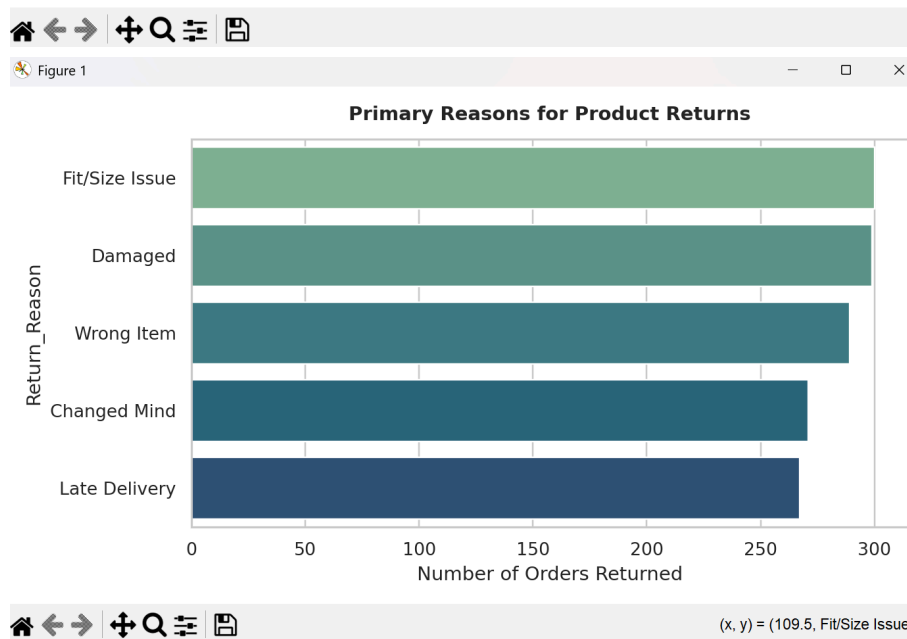
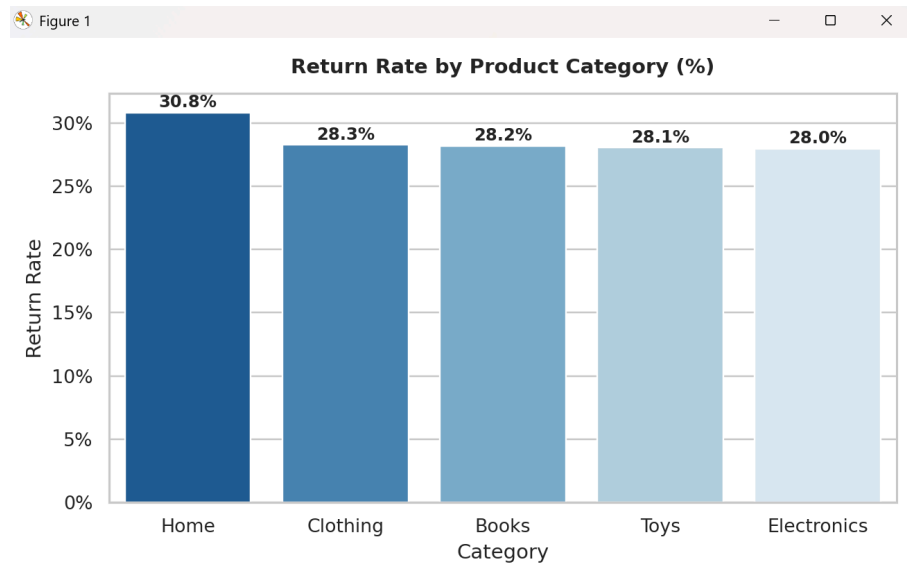
10.7 Payment Method Analysis

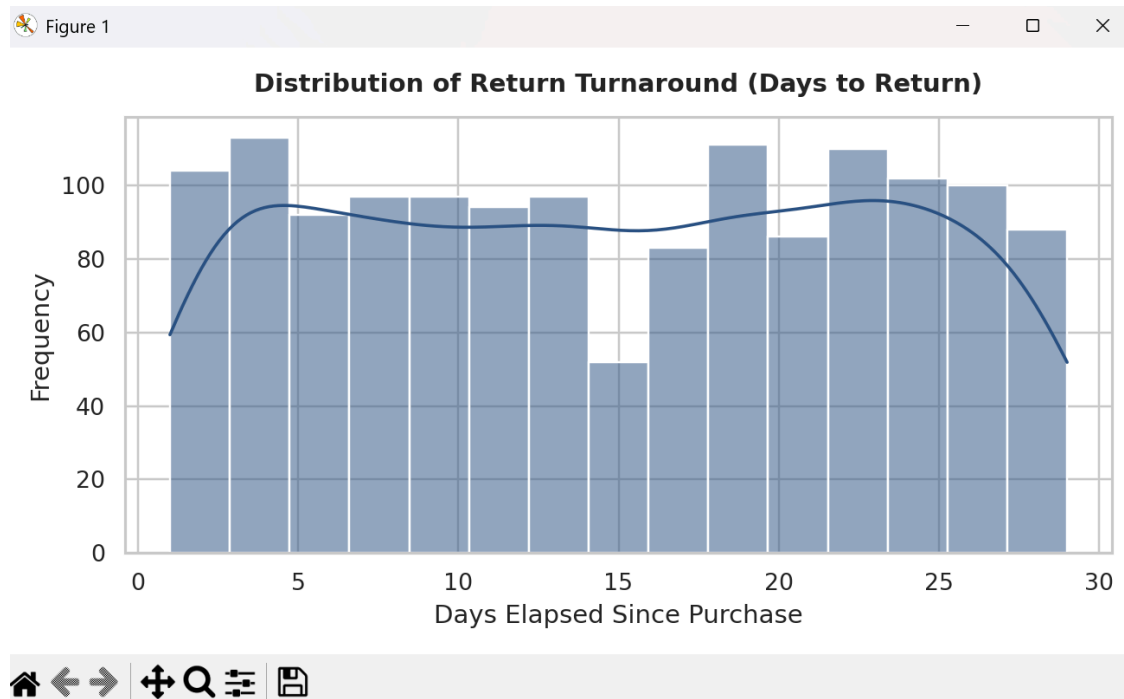
Return rates across payment methods are analyzed to identify possible behavioral patterns.

10.8 Financial Analysis

Return costs and profit/loss are analyzed to determine the financial impact of returns.

Products with both high return rates and high return costs should receive priority for intervention.





11. Statistical Analysis

```

IDLE Shell 3.11.9
File Edit Shell Debug Options Window Help
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr 2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Siri Chand/OneDrive/Desktop/ba_miniproject.py
[*] 'returns_sustainability_dataset.csv' not found. Initializing dataset with Kaggle schema...
=====
SECTION 1: EXPLORATORY DATA ANALYSIS & SUMMARY METRICS
=====
Total Transactions Analyzed      : 5,000
Total Products Returned         : 1,304
Overall Product Return Rate     : 26.08%
Cumulative Reverse Logistics Cost: $23,683.93
Net Profit / Loss Impact       : $35,967.20

--- Summary Statistics (Numerical Predictors) ---
              mean      std      min      median      max
Product_Price 75.366656 61.153177 15.009465 55.863737 584.742973
Order_Quantity 1.570000 0.882075 1.000000 1.000000 4.000000
Discount_Applied 0.226660 0.175492 0.000000 0.200000 0.500000
User_Age      42.552200 14.277410 18.000000 43.000000 67.000000

--- Return Rate & Financial Impact by Product Category ---
              Total_Orders  Returned_Orders  Return_Rate_Pct  Total_Return_Cost  Avg_Days_to_Return
Product_Category
Clothing          1758          1023          58.19          18678.68          14.82
Electronics       1341          124           9.25          2214.52          14.95
Home & Kitchen     724           63           8.70          1146.23          14.35
Books             710           58           8.17          1002.73          14.10
Toys              467           36           7.71           641.77          15.14

--- Primary Return Reasons Distribution ---
              Count  Percentage  Cum_Percentage
Return_Reason
Damaged        279          21.40          21.40
Changed Mind   277          21.24          42.64
Wrong Item     262          20.09          62.73
Fit/Size Issue 254          19.48          82.21
Late Delivery  232          17.79          100.00

IDLE Shell 3.11.9
File Edit Shell Debug Options Window Help

--- Return Rate by Shipping Method ---
              Orders  Return_Rate
Shipping_Method
Express          1232          26.14%
Next-Day         789          34.35%
Standard         2979          23.87%

Chi-Square Test (Category vs Return): Chi2 = 1450.71, p-value = 0.0000e+00 (Significant: True)
=====
SECTION 2: MODEL TRAINING & CROSS-VALIDATION
=====

--- 5-Fold Stratified Cross-Validation Results ---
              Model      CV Accuracy      CV Precision      CV Recall      CV F1-Score      CV ROC-AUC
Logistic Regression 0.8320 (±0.013) 0.7072 (±0.029) 0.6079 (±0.023) 0.6537 (±0.024) 0.8687 (±0.015)
Random Forest       0.8417 (±0.006) 0.7360 (±0.021) 0.6146 (±0.018) 0.6694 (±0.010) 0.8722 (±0.015)
Gradient Boosting   0.8420 (±0.007) 0.7314 (±0.021) 0.6242 (±0.020) 0.6732 (±0.013) 0.8755 (±0.015)

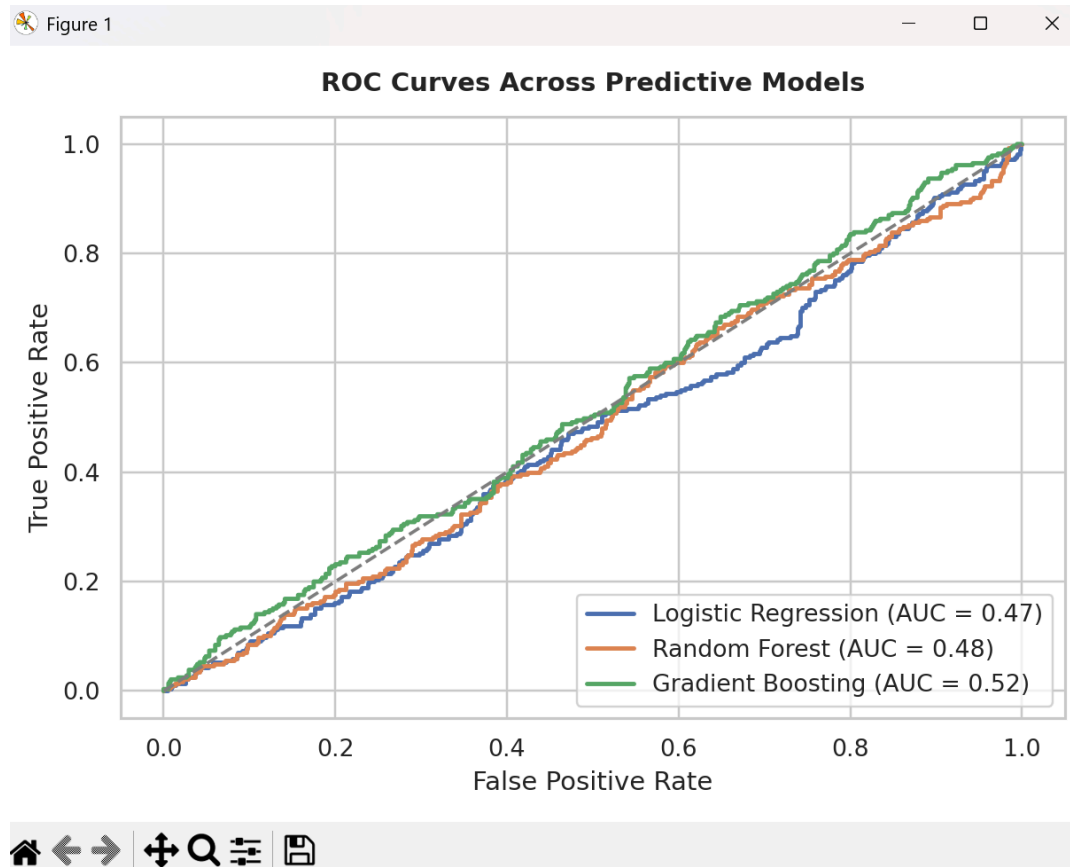
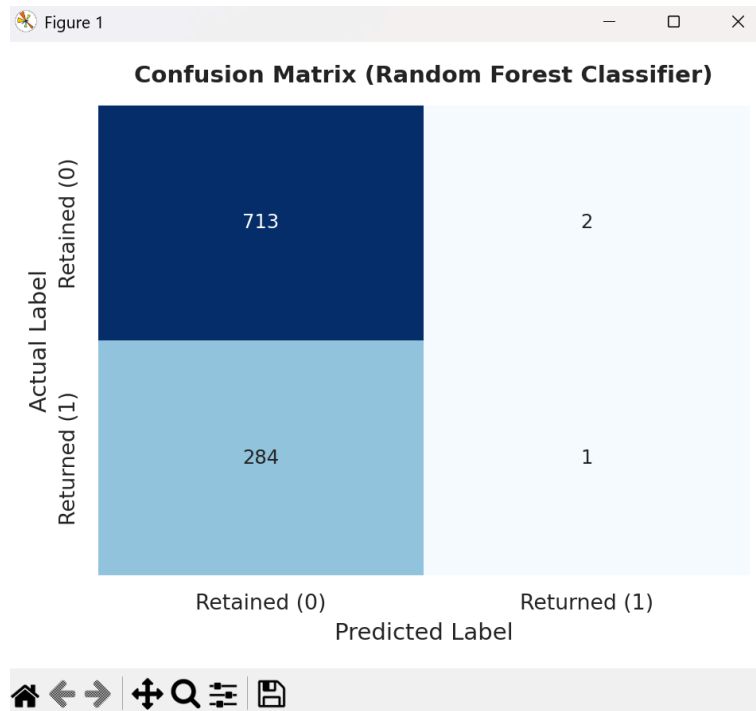
=====
SECTION 3: HOLDOUT TEST SET EVALUATION
=====

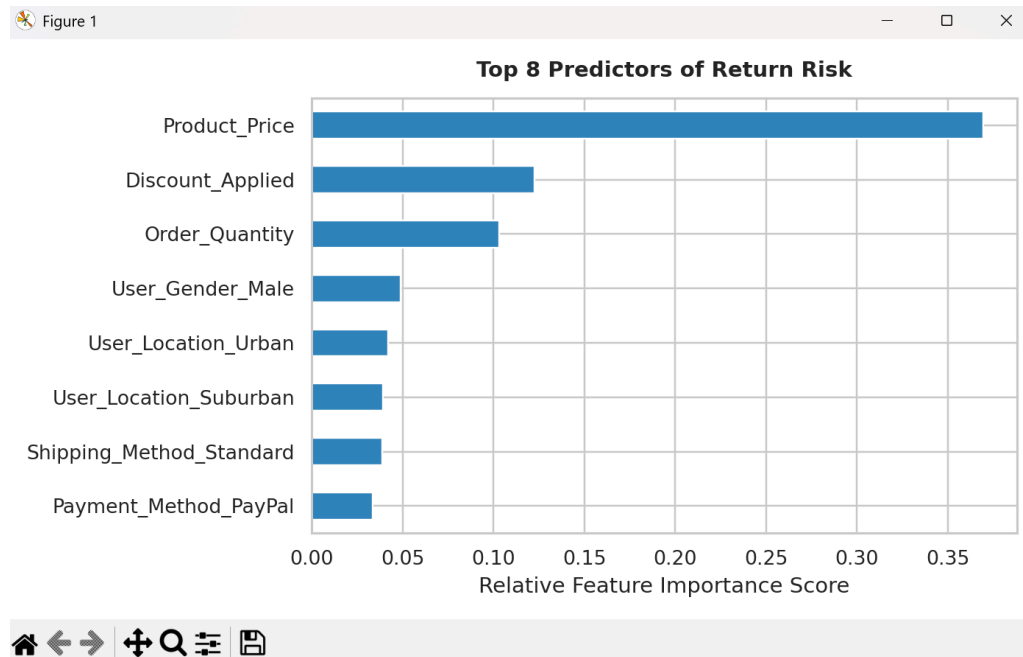
--- Model Performance Comparison (Test Split) ---
              Model  Accuracy  Precision  Recall (Sensitivity)  Specificity  F1-Score  ROC-AUC
Random Forest    0.846    0.7512    0.6130    0.9283    0.6751    0.8698
Logistic Regression 0.840    0.7285    0.6169    0.9188    0.6680    0.8698
Gradient Boosting 0.837    0.7188    0.6169    0.9147    0.6639    0.8697

--- Confusion Matrix Raw Counts (TP, FP, TN, FN) ---
              Model  TP  FP  TN  FN
Random Forest    160  53  686  101
Logistic Regression 161  60  679  100
Gradient Boosting 161  63  676  100

```

12. Data Visualization





13. Business Analytics Technique / Model

The project combines four types of analytics.

13.1 Descriptive Analytics

Descriptive analytics answers:

What happened?

Examples:

- Total orders.
- Total returns.
- Return rate.
- Common return reasons.
- Average return duration.

13.2 Diagnostic Analytics

Diagnostic analytics answers:

Why did it happen?

The analysis investigates relationships between return behavior and:

- Product category.
- Product price.
- Discounts.
- Shipping method.
- Payment method.
- Customer characteristics.

13.3 Predictive Analytics

Predictive analytics answers:

What is likely to happen?

Machine-learning models are used to predict whether an order is likely to be returned.

Models considered include:

- Logistic Regression.
- Decision Tree.
- Random Forest.
- Gradient Boosting.

13.4 Prescriptive Analytics

Prescriptive analytics answers:

What should the business do?

The results are converted into recommendations for:

- Product design.
- Product information.
- Quality control.
- Packaging.
- Return policy.
- Reverse logistics.

14. **Model Development / Analysis**

The predictive modeling process follows these steps:

Data



Cleaning



Feature Engineering



Feature Selection



Encoding



Train/Test Split



Model Training



Prediction



Evaluation



Best Model Selection

The dataset is divided into training and testing sets.

A typical split is:

- **80% Training**
- **20% Testing**

The target variable is:

Return_Status

Potential features include:

Product Category

Product Price

Order Quantity

Discount Applied

User Age

User Gender

User Location

Payment Method

Shipping Method

15. Model Evaluation / Validation

```

--- Detailed Classification Report: Random Forest ---
              precision    recall  f1-score   support

Retained (0)       0.87      0.93      0.90       739
Returned (1)       0.75      0.61      0.68       261

   accuracy              0.85       1000
  macro avg       0.81      0.77      0.79       1000
 weighted avg       0.84      0.85      0.84       1000

--- Top Predictive Feature Importances ---
Product_Category_Clothing      0.4758
Product_Price                  0.1477
Discount_Applied               0.1396
Product_Category_Electronics   0.0612
Product_Category_Home & Kitchen 0.0331
Order_Quantity                 0.0239
Shipping_Method_Next-Day       0.0200
Product_Category_Toys          0.0192
>>>

```

16. Business Insights

The analysis is expected to provide insights into:

Return Drivers

Identify the main factors associated with returns.

High-Return Categories

Identify product categories with above-average return rates.

Return Reasons

Identify the most frequent return reasons.

Return Timing

Determine typical return timing and whether early or late returns dominate.

Financial Impact

Identify products or categories with high return costs and negative financial impact.

Predictive Factors

Identify the features that contribute most to the predictive model.

Operational Factors

Analyze whether shipping and payment methods show differences in return behavior.

17. Recommendations / Decision-Making

17.1 Product Design

If defects or quality issues are common:

- Improve quality-control processes.
- Redesign frequently returned products.
- Improve product durability.
- Conduct additional pre-shipment testing.

17.2 Product Information

If products are frequently returned because they are not as expected:

- Improve product descriptions.
- Add detailed specifications.
- Provide accurate dimensions.
- Add additional product images.
- Include customer reviews and FAQs.

17.3 Packaging

If damage-related returns are significant:

- Improve packaging materials.
- Introduce stronger protective packaging.
- Analyze transportation damage.
- Review logistics partners.

17.4 Return Policy

Return policies should not simply be made stricter.

Instead, businesses should identify **avoidable returns** and address their causes.

Possible actions include:

- Better return instructions.
- Exchange options.

- Product-specific return policies.
- Monitoring repeated return patterns.
- Using return-risk predictions for operational planning.

17.5 Reverse Logistics

Businesses can improve reverse logistics by:

- Consolidating return shipments.
- Optimizing return transportation.
- Establishing efficient inspection procedures.
- Restocking suitable products quickly.
- Refurbishing products where economically viable.
- Reselling eligible returned products.
- Recycling unsuitable products.

18. Implementation / Dashboard

Power BI is used as the primary business-intelligence and visualization tool.

The dashboard consists of four main pages.

Page 1: Executive Overview

KPI Cards

- Total Orders.
- Total Returns.
- Return Rate.
- Average Days to Return.
- Total Return Cost.
- Total Profit/Loss.

Charts

- Return Status.
- Returns by Product Category.
- Returns by Return Reason.
- Monthly Return Trend.

Business Question

What is happening with product returns?

Page 2: Return Analysis

Visualizations

- Return Rate by Product Category.
- Return Reasons.
- Days-to-Return Distribution.
- Discount vs Return Rate.
- Shipping Method vs Return Rate.
- Payment Method vs Return Rate.

Business Question

Why are products being returned?

Page 3: Reverse Logistics & Financial Impact

Visualizations

- Return Cost by Product Category.
- Return Cost by Return Reason.
- Profit/Loss by Return Status.
- Average Return Cost.
- Waste Avoided.
- CO₂ Saved.

Business Question

What is the operational and financial impact of returns?

Page 4: Predictive Analytics

The machine-learning predictions generated in Python are imported into Power BI.

The page displays:

- Predicted Return Probability.
- High-Risk Orders.
- Medium-Risk Orders.
- Low-Risk Orders.

- Model Performance.
- Feature Importance.

Example:

Order ID	Return Probability	Risk
10001	82%	High
10002	24%	Low
10003	58%	Medium

The exact risk thresholds will be finalized based on model performance and business requirements.

Business Question

Which orders are most likely to be returned?

19. Results and Discussion

The results section will combine findings from Python and Power BI.

The final project will report:

- Overall return rate.
- Most common return reasons.
- Highest-returning product categories.
- Average time to return.
- Relationship between discounts and returns.
- Relationship between product price and returns.
- Differences across shipping methods.
- Differences across payment methods.
- Total return costs.
- Financial impact.
- Statistically significant relationships.
- Best machine-learning model.
- Model accuracy and ROC-AUC.
- Most important predictive features.

The Power BI dashboard will present these results in an interactive format.

The findings will then be interpreted from a business perspective.

For example, if a particular product category has a high return rate and the primary reason is product quality, the recommendation would be to investigate product design and quality control rather than simply tightening the return policy.

20. Limitations

The project has several limitations.

1. The dataset is synthetic rather than actual company data.
2. Synthetic data may not fully represent real customer behavior.
3. Some important real-world variables may not be available.
4. Return reasons may be simplified.
5. Historical relationships may change over time.
6. Machine-learning models identify patterns and associations but do not necessarily establish causality.
7. The predicted return probability depends on the quality and representativeness of the available data.

21. Future Enhancements

Future versions could include:

Real-world company return data.

Customer review analysis using NLP.

Text analysis of return reasons.

Customer-level return prediction.

Product-level return forecasting.

Return-cost prediction.

Real-time return-risk scoring.

Inventory optimization.

Reverse-logistics route optimization.

XGBoost and advanced machine-learning models.

Live Power BI or Streamlit dashboard.

22. Conclusion

This project uses data analytics and machine learning to understand and predict e-commerce product returns.

The analysis focuses on return reasons, product characteristics, purchase timing, discounts, shipping methods, payment methods, and financial impact.

By identifying the factors associated with returns, businesses can take proactive measures such as improving product design, product information, packaging, quality control, and return policies.

The project demonstrates how **descriptive, diagnostic, predictive, and prescriptive analytics** can be combined to improve return management and reverse-logistics efficiency.

The ultimate goal is to reduce avoidable returns while maintaining customer satisfaction and improving business profitability.

23. Tools and Technologies Used

Tool / Technology	Purpose
Python	Data analysis and machine learning
Pandas	Data cleaning and manipulation
NumPy	Numerical analysis
Matplotlib	Data visualization
Seaborn	Exploratory data visualization
Scikit-learn	Machine learning and model evaluation
SciPy	Statistical analysis
Power BI	Interactive dashboard and business reporting
Jupyter Notebook / Google Colab	Python development and analysis
Kaggle	Dataset source
Git & GitHub	Version control and project hosting
Power BI Dashboard	

Power BI will be used to create an interactive dashboard containing:

- Total Orders
- Total Returns
- Return Rate
- Average Days to Return
- Total Return Cost
- Profit/Loss
- Return Reasons
- Return Rate by Product Category
- Return Rate by Shipping Method
- Return Rate by Payment Method
- Monthly Return Trends
- Return Cost Analysis
- Product Return Risk

The dashboard will provide an easy-to-understand view of the key business insights and support data-driven decision-making.

24. Project / GitHub Link

<https://github.com/siri-cn/BUSINESS-ANALYTICS-MINI-PROJECT-2117240030143.git>

25. References

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